

A Reachability-Pruned Activity-Based Demand Model for MATSim: Congestion Response and Demand Generation in Higashi-Hiroshima

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I. Motivation

The behavioural realism of an agent-based simulation ultimately depends on the quality of its demand representation. Since detailed activity schedules are rarely available for entire populations, MATSim demand pipelines often rely on statistical matching to assign survey-based mobility patterns to synthetic agents. This limits the behavioural diversity represented in the demand relative to real-world travel [1]. Activity-based dynamic discrete choice models (DDCMs) instead solve each person's day from their own constraints under a utility model, making variability structural rather than bounded by a reference sample. Yet these models are typically estimated against exogenously specified travel costs, rather than the time-varying network conditions generated by the agent-based simulator itself.

This paper asks two questions: does the coupling matter, in the sense that simulated congestion improves the model's fit to observed activity-travel diaries (Section V); and does it scale, in the sense that the estimated model can generate demand that MATSim can load and assign for an entire study area (Section VI)?

II. The Demand Model

The model is a dynamic discrete choice model of a full activity-travel day [2]. At each discrete time step a person continues the current activity or travels, choosing among the study area's zones and the available modes so as to maximise the expected utility of the rest of the day; preferences are estimated from observed diaries by a nested fixed point algorithm, with backward induction performed inside each likelihood evaluation. Solving that exactly is combinatorially large, so the state space is restricted by a space-time prism: a state is kept only if the day can still reach it and still return home before the horizon. The discarded states are ones no feasible day could occupy, so the value function and its gradient are unchanged and the induction stays exact. That technique is not this paper's contribution but a companion one's; it matters here only as what makes the model tractable at a real study area's scale [3].

The travel cost entering that value function is ordinarily a fixed origin-destination table, and replacing it with one that varies by departure time is what couples the model to a traffic simulator. That table enters the transition function as well as the utility, so a congested one does not merely make an alternative worth less: it moves the arrival state, and can push a state beyond the prism altogether. Congestion can therefore change the choice set itself, which the coupling makes testable rather than assumes.

III. Coupling Method

The literature couples demand and supply models by one-way handoff, by an iterated fixed point between demand and the level-of-service the simulator generates from that model's own output [4, 5], or at the scoring-function level. In this study, information is exchanged once in each direction rather than iterated to a fixed point.

A synthetic population built independently of the demand model, and not generated by

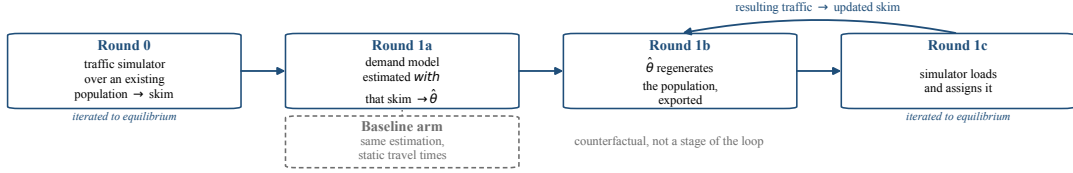


Figure 1: The coupling loop as it would run in full; the baseline arm sits off it.

it, is simulated through MATSim’s co-evolutionary loop over route, mode and departure time, run until agents approximate a stochastic user equilibrium [1], and a time-varying skim is extracted from the resulting traffic by mode and departure-time bin. That skim replaces the fixed lookup inside the value function without approximation, and the model is re-estimated against it. The estimated parameters are then held fixed and used once to regenerate full-day plans, exported with person identity, home location and scoring-relevant attributes unchanged; MATSim still runs its own replanning on top, so what is supplied is a starting point, not a replacement for it.

Nothing from the generation step is fed back into estimation, and the first question requires exactly that: the estimation skim comes from a population the simulator already had, so it is exogenous to the parameters estimated against it, where one taken from the population generated here would be a function of them. This also fixes what may iterate. Estimated preferences describe people rather than the network they moved through, so they are recovered once and held fixed; what repeats is regeneration at those parameters, assignment and an updated skim, the return Figure 1 draws. One pass of it is reported here, each further pass being a full simulator run and a full regeneration.

IV. Empirical Setting

The study area is Higashi-Hiroshima, represented by 144 zones, with the day in 15-minute steps and five modes available. Preferences are estimated on 7,467 person-days from the local travel diary survey, in 972 evaluation groups. The population carried through the loop is a synthetic population of 214,216 persons built for the same area independently of this project; simulating it for 100 MATSim iterations yields the skim, which covers all 19,462 non-diagonal ordered pairs and varies for car only (Section VII). Time-of-day variation is muted on average but genuinely right-skewed: against a 22:00–06:00 off-peak, the peak ratio averages 1.024 in both the 07:00–10:00 and the 16:00–19:00 window, with p90 of 1.092 and 1.078. The tail carries the signal and is directionally structured: the worst pairs reach $3.85\times$ in the AM and $6.53\times$ in the PM, and the two tails are approximate mirror images, seven of the ten worst AM pairs sharing an *origin* zone against six of the ten worst PM pairs sharing a *destination*, which is what a concentrated employment location produces.

V. Does the Model Register the Congestion?

The cleanest check holds everything fixed except the travel-cost table: one parameter vector, evaluated twice over the same persons and groups, once against each table, with no optimiser in between, so the comparison does not depend on either estimation having converged. The vector held fixed is the optimum obtained under the *static* table, which is deliberately unfavourable to the skim. Nevertheless, the static table yields a lower log-likelihood (Table 1), by 222 points at an identical parameter count, equivalent to 0.030 per person.

Two qualifications belong with this. The skim is itself simulated, so the result measures agreement with the simulator’s reconstruction of congestion rather than with observed traffic; and the two evaluations score slightly different numbers of decision steps (406,381 against 406,270) because a few paths change feasibility with the table, making this a comparison of two descriptions of the environment rather than a nested test.

That last difference is the choice-set channel of Section II becoming measurable, and Table 1 sizes it. Decomposing the day’s first-state log-sum separates it from the cost of trips merely taking longer; it reaches 1,350 of the 1,364 workers, so it is pervasive, but it

Table 1: One parameter vector, held fixed, under both travel-cost tables.

	static OD	MATSim skim	change
diary log-likelihood, 7,467 persons	−170,897.20	−170,674.85	+222.35
reachable states, 144 zones	866,220	865,038	−0.14%
reachable transitions	89,164,093	87,301,511	−2.09%
log-sum welfare per worker ($n = 1,364$)		mean ΔV_0	share
trips taking longer, choice set unchanged		−1.128	99.9%
alternatives lost outright		−0.001	0.1%
total		−1.129	100%

carries a thousandth of the loss. Congestion is felt here as delay, not as foreclosure.

VI. Generating Demand

All 214,216 persons map into the state space with no coordinate fallback, and reduce to four distinct graph topologies regardless of population size, because the attributes that vary most across persons affect utility rather than the graph, so those graphs cost the same for the full population as for two hundred. Holding the estimated parameters fixed, full-day plans are regenerated and exported, and MATSim loads and assigns the result over 100 iterations. Trip generation matches the source population closely, 2.66 trips per person against 2.66. Mode-level results are still being analysed and will be presented at the conference. This is a capability and mechanism demonstration, not a production-scale realism claim.

VII. Limitations and Future Work

The skim is car-only, so the congestion response is only partially exercised. Rail is separately limited in the origin–destination data rather than in the model: service is present on 270 of 20,736 zone pairs, against an observed rail mode share of 23.1%. A mode available on 1.3% of pairs cannot reproduce that share whatever the preferences are, and any comparison of mode composition inherits the ceiling.

The two models’ choices overlap, though more narrowly than such a coupling might suggest. MATSim models neither the activity sequence nor participation in an activity as such [6], and the configuration used here innovates on route, departure time and mode without a destination-choice operator, so the sequence and locations the demand model produces survive replanning intact while timing and mode are re-decided, an overlap not reconciled here. Neither estimation arm is claimed as a maximum-likelihood point either: the static one stopped by a practical-convergence rule at $\|\nabla\|_\infty = 3.4 \times 10^{-3}$, the arm carrying the skim at a wall-clock limit after five accepted steps, at 0.54. The comparison above does not require one.

Closing the loop, finally, needs compute rather than a change of method, since the parameters stay fixed and only generation and assignment repeat. Even the closest prior studies [4, 5] report neither their iteration count nor their convergence time.

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